

ARUN KUMAR SOORA

☎ (317) 508-6885 ✉ arunkumarsoora@gmail.com 🔗 [linkedin.com/in/arun-kumar-soora](https://www.linkedin.com/in/arun-kumar-soora) 🌐 github.com/ArunSoora

Education

Indiana University Indianapolis

Master of Science in Health Informatics, GPA: 3.55/4.0

Aug 2023 - May 2025

Indianapolis, Indiana

Gurunanak Institute Of Technical Campus

Doctor of Pharmacy, GPA: 3.4/4

Jul 2016 - Aug 2022

Hyderabad, Telangana, India

Experience

Indiana University –School of Health and Human Sciences

Jan 2024 - Present

Clinical Research Coordinator

Indianapolis, Indiana

- Streamlined real-time patient data collection using Actigraph GT3X, ActiLife, and Qualtrics eCRFs for 154 participants; leveraged Excel, Python, and SQL for data cleaning, analysis, and dashboard reporting.
- Ensured GCP and IRB compliance through consent tracking, deviation logs, and audit-ready documentation; reduced manual errors by 30% with automated data pipelines and quality control checks.

GTechnologies PTY LTD

Jan 2025 - May 2025

Data Analyst Intern

Sydney, Australia

- Developed a cloud-based EMR-style tool using React.js, Python, and SQL with custom clinical forms and workflow automation to support HIPAA-compliant data capture.
- Analyzed patient data using Excel and SQL; built Power BI dashboards to support care planning, outcome tracking, and data-driven decision-making.

Indiana University, BioHealth Informatics Research Center

Sep 2023 - May 2025

Graduate Research Assistant

Indianapolis, Indiana

- Integrated 500+ multi-source health datasets into a centralized research platform, advancing global health informatics research.
- Utilized SQL, Excel, and Power BI to collect, clean, analyze, and visualize national and Serbian health data; supported faculty projects with interactive dashboards for real-time insights.

Care Hospitals

Jul 2022 - Jun 2023

Clinical Research Coordinator – Cardiology & Neurology

Hyderabad, India

- Reviewed and edited clinical documentation across cardiology and neurology, ensuring accuracy, HIPAA/GCP compliance, and maintaining Excel-based logs to support EHR workflows and patient tracking.
- Managed 5,000+ patient records using Excel for outcome monitoring, delivered 200+ drug interaction reviews, 100+ ADR reports, and provided counseling to improve adherence and support regulatory reporting.

Skills

Data Analysis & Visualization: SQL, Python, R, Tableau, Power BI, Excel, Pyplot, ggplot2

Statistical Methods: Hypothesis Testing, Regression Analysis, Time-Series Analysis, Sentiment Analysis

Machine Learning & NLP: Scikit-learn, TensorFlow, Numpy, Pandas, Natural Language Processing (NLP), Clustering

Data Engineering & ETL: Talend Open Studio, SQL, Docker, REST APIs, JSON, XML

EHR systems: OpenEMR, OpenMRS, Cerner, SMART-on-FHIR

Healthcare Standards: SNOMED, ICD, LOINC, CPT, RxNorm, HL7, FHIR, OHDSI, OMOP CDM

Tools & Platforms: DHIS2, GitHub, Postman, Oxygen XML Editor, REDCap, Qualtrics, Jupyter, RStudio, Lucidchart

Projects

VR Cognitive Performance and Stroke Rehabilitation Data Analysis | Python, Excel

Jan 2025 - May 2025

- Analyzed cognitive function data in VR interactions, applying Spearman correlation and APA standards.
- Evaluated stroke rehabilitation effectiveness by comparing accelerometer-generated M2 scores with traditional assessment metrics (FMA, MAL) using linear regression and scatter plots in PyCharm.

Project Sprint, Achieving Interoperability | XML, Postman, JSON, API

Jan 2024 - May 2024

- Mapped clinical data to standardized codes (ICD-10, SNOMED, LOINC, RxNorm) for a hypertensive urgency use case, ensuring HL7-FHIR compliance and improving workflow clarity through UML/BPMN modeling.
- Developed Built and transformed HL7 CDA documents (XML to JSON) using XSLT; enabled real-time data exchange between OpenMRS and OpenELIS through API testing and validation in Postman.

Healthcare Insurance Cost Prediction Using Regression Modeling | R

Jan 2024 - May 2024

- Built a multiple linear regression model in R (78% accuracy) to predict healthcare insurance costs using factors like BMI, age, and smoking status; enhanced model reliability through data cleaning and feature selection.
- Performed EDA and created clinical visualizations using ggplot2 to identify cost-related risk factors, supporting patient stratification and evidence-based resource planning.

Cervical Cancer Epidemiology | *SQL, Python, Machine Learning*

Aug 2023 - Dec 2023

- Built predictive models (Logistic Regression, SVM, Random Forest, Decision Tree) using Python and SMOTE to identify high-risk cervical cancer cases from 858-patient dataset, achieving up to accuracy and strong AUC scores (0.87–0.93); conducted statistical tests (Chi-square, ANOVA, t-test) to validate key risk factors (age, smoking, contraceptives, STDs).
- Cleaned and transformed clinical data (median/binary imputation), performed EDA and correlation analysis, and visualized insights using Seaborn; improved data quality and model reliability while supporting personalized screening strategies and evidence-based cancer prevention.

A Prospective Study: Prevention and Treatment of Anemia | *Python, SQL*

Jun 2020 - Sep 2021

- Conducted a prospective study on iron-deficiency anemia in 100+ heart failure patients using lab diagnostics and echocardiography to assess prevalence and severity.
- Recommended IV iron therapy protocols, improving 30-day medication adherence, reducing readmissions, and supporting data-driven patient management.

Relevant Publications

- * Students consider the impact of Democratic and Republican Healthcare Policies on Health Informatics and Health Information Technology